Responsible AI and Model Interpretability

Naeemullah Khan
naeemullah.khan@kaust.edu.sa

KAUST Academy
King Abdullah University of Science and Technology

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Learning Outcomes

- Identify where bias enters an ML system — data collection, labels, proxy variables, historical patterns, and feedback loops — and audit a model with disaggregated metrics.
- State the main group fairness definitions and explain why they cannot all hold at the same time.
- Apply SHAP and LIME to a trained model, and describe their assumptions and failure modes.
- Write a model card and a datasheet for a model you are shipping.
- Summarise what current regulation and frameworks (EU AI Act, NIST AI RMF) ask of a practitioner.

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Where This Lecture Sits

So far the course has optimised a single number: accuracy, F1, RMSE. That number is computed **over the whole test set**.

The gap this lecture closes

A model that is accurate *on average* can still be systematically wrong for a particular group of people, and can be right for reasons you would not accept if you could see them.

- **Part 1 — Fairness:** how to find and reduce systematic error across groups.
- **Part 2 — Interpretability:** how to find out what a model is actually using.
- **Part 3 — Documentation and governance:** how to write it down and who checks it.

Not covered here: agent- and LLM-specific safety — prompt injection, tool misuse, excessive agency — which is the subject of a dedicated lecture in **the NLP course**.

We start with four documented cases. Each one failed in a different place in the pipeline.

Case 1: COMPAS Recidivism Scores (2016)

The system. COMPAS is a commercial risk-assessment tool (Northpointe, now Equivant) that scores a criminal defendant's likelihood of re-offending on a 1–10 scale. US courts have used such scores to inform bail, sentencing, and parole decisions.

The analysis. ProPublica journalists obtained scores for more than 7,000 people arrested in Broward County, Florida in 2013–2014 and checked, two years later, who actually re-offended.

	White	**Black**
Labelled higher risk, but did *not* re-offend	23.5%	44.9%
Labelled lower risk, yet *did* re-offend	47.7%	28.0%

- Overall, 61% of those scored likely to re-offend were arrested for some crime within two years.
- Of those predicted to commit a *violent* crime, 20% did.

J. Angwin, J. Larson, S. Mattu and L. Kirchner, "Machine Bias," *ProPublica*, 23 May 2016, propublica.org.

Case 1: The Disagreement That Followed

Northpointe replied that its score was **equally accurate for both groups**: among defendants given the same score, roughly the same fraction re-offended regardless of race.

Both sides were reporting true numbers

ProPublica measured **error rates** conditioned on the true outcome (who got wrongly flagged). Northpointe measured **calibration** conditioned on the score (what a score means). These are different quantities.

- Chouldechova (2017) and Kleinberg, Mullainathan and Raghavan (2016) showed independently that when the two groups have **different base rates** of the outcome, a score cannot satisfy both criteria at once, except in degenerate cases.
- So this was not a bug that better engineering would fix. It was a **choice between incompatible definitions of fair** — and nobody had made it explicitly.

We return to this result in Section 4.

Case 2: A Resume-Screening Tool (Reuters reporting, 2018)

Note on evidence

This case comes from **news reporting based on anonymous sources**, not from a published study. Treat the mechanism as illustrative and the details as journalistic.

According to Reuters, a team at Amazon built experimental tools from 2014 that scored job applicants one to five stars. By 2015 the team found the system was not rating candidates for software-developer roles in a gender-neutral way.

- The models were trained on resumes submitted to the company over roughly a decade — a pool that was predominantly male in technical roles.
- The reporting states the system penalised resumes containing the word "women's" and downgraded graduates of two all-women's colleges.

- Amazon said the tool was never used by recruiters to evaluate candidates, and the project was abandoned.

The mechanism, stated plainly

The label was not “a good engineer.” It was “a person this company hired before.” Removing the gender field does not help, because the model finds **proxies**. The learning worked; the **target definition** was the defect.

J. Dastin, “Amazon scraps secret AI recruiting tool that showed bias against women,” *Reuters*, 10 October 2018.

Case 3: Gender Shades — Disaggregated Error Rates (2018)

Buolamwini and Gebru audited three **commercial** gender-classification APIs on the *Pilot Parliaments Benchmark* (PPB): 1,270 parliamentarians from three African and three European countries, labelled by Fitzpatrick skin type and by binary gender.

Classifier	All	Darker female	Darker male	Lighter female	Lighter male
Microsoft	6.3%	20.8%	6.0%	1.7%	0.0%
Face++	10.0%	34.5%	0.7%	9.8%	0.8%
IBM	12.1%	34.7%	12.0%	7.1%	0.3%

Error rate = 1− true positive rate, evaluated April–May 2017.

- The aggregate column looks acceptable. The intersectional columns do not.
- Marginal slices hide it: an 8.1%–20.6% gap by gender alone, 11.8%–19.2% by skin type alone.
- Maximum reported error-rate gap between the best- and worst-classified subgroups: **34.4%**.

Two lessons. (1) Slice by every group *and by combinations*, and report the worst slice with its sample size. (2) Quote the study for what it measured — binary gender classification, one benchmark, three APIs, one point in time — not as "face recognition does not work."

J. Buolamwini and T. Gebru, "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification," *PMLR* 81:77–91, FAT* 2018, Table 4 and Section 4.1.

Case 4: A Health Risk Algorithm and Its Proxy Label (2019)

The system. A widely deployed algorithm identified patients who should be enrolled in high-intensity care-management programmes. It predicted **future health-care costs** as a stand-in for **future health needs**.

The defect

Less money is spent on Black patients at the same level of illness. Predicting cost therefore predicts *spending*, not *sickness* — and the algorithm concluded that equally sick Black patients were healthier.

- At a given risk score, Black patients were measurably sicker (more uncontrolled chronic conditions).
- Reformulating the label removed most of the disparity: the share of Black patients flagged for extra help rose from **17.7% to 46.5%**.
- The sensitive attribute was **not** an input. The **label** carried the bias.

Case 4: A Health Risk Algorithm and Its Proxy Label (2019) (cont.)

Z. Obermeyer, B. Powers, C. Vogeli and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science* 366(6464):447–453, 2019.

What the Four Cases Have in Common

Case	Where it broke	What would have caught it
COMPAS	Conflicting fairness criteria, never chosen explicitly	Naming the fairness criterion before deployment
Resume screening	Target definition (“past hires”) and proxies	Asking what the label actually measures
Gender Shades	Benchmark composition and unreported slices	Disaggregated, intersectional evaluation
Health risk score	Proxy label (cost for need)	Auditing the label against the real outcome

None of these were caused by a bad optimiser or a small test set. All four were **specification** problems that a standard evaluation report would not surface.

First: "Bias" Means Three Different Things

Statistical bias

$\mathbb{E}[\hat{\theta}] - \theta \neq 0$.

The estimator is systematically off. Neutral term; half of the bias–variance trade-off.

Inductive bias

The assumptions a model class makes.

Convolutions assume locality; linear models assume additivity. **Necessary** — without it there is no generalisation.

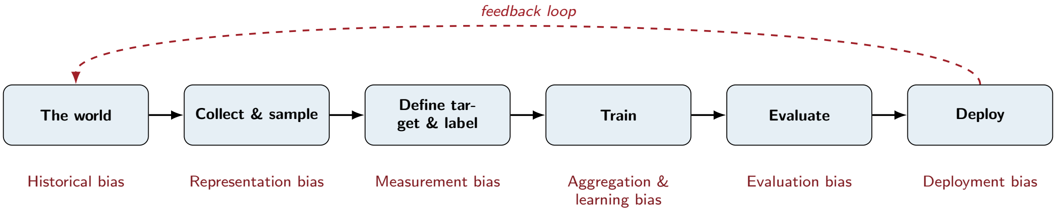
Societal / harmful bias

Systematically different quality of service or outcome across groups of people.

This lecture's subject.

The three are unrelated. A model can be statistically unbiased and still harmful; a strong inductive bias can either mask or expose a societal one.

Bias Enters at Every Stage, Not Just “The Data”



Taxonomy after H. Suresh and J. V. Guttag, “A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle,” EAAMO 2021, arXiv:1901.10002v5.

Historical and Representation Bias

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Historical bias

Present even with perfect sampling and perfect measurement, because the world the data records is itself unequal.

- An image search for "CEO" that reflects who currently holds the job.
- A loan model trained on approvals from an era of discriminatory lending.

More data does not fix this. It is a question about the objective, not the sample.

Representation bias

The development sample under-represents part of the use population, so the model generalises poorly there.

- A dermatology model trained mostly on light skin.
- A speech model trained on one accent.
- Gender Shades: existing face benchmarks were 79.6% and 86.2% lighter-skinned subjects.

More *targeted* data does help — but you must first know the composition of your data.

Measurement Bias: The Label Is a Proxy

You almost never observe the thing you care about. You observe something correlated with it.

What you want	What you record	What the gap encodes
Health need	Health-care spending	Unequal access to care
Committing a crime	Being arrested	Unequal policing intensity
Job performance	Was hired / was promoted	Past hiring decisions
Creditworthiness	Repaid a loan they were *granted*	Who was offered credit
Content quality	Clicks and watch time	Engagement, not quality

The one question to ask on every project

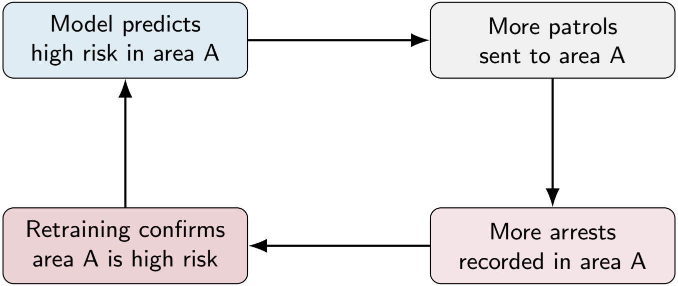
"My label is X. What exactly had to happen in the world for X to be recorded, and does that differ by group?"

Aggregation, Learning, Evaluation and Deployment Bias

- **Aggregation bias.** One model for populations that genuinely behave differently. Suresh and Guttag point to medicine: patients of different sexes with similar underlying conditions can present and progress differently, so one model is optimal for neither.
- **Learning bias.** The modelling choice itself amplifies a disparity. Optimising average loss lets a large majority group dominate the gradient; aggressive compression or regularisation tends to cost the tails most.
- **Evaluation bias.** The benchmark does not represent the use population, so the reported number is not the number you will get. This is exactly the Gender Shades finding about existing face benchmarks.
- **Deployment bias.** The model is used differently from how it was framed. A risk score built to *rank* people for services becomes a *decision* about their liberty; a triage aid becomes an autopilot.

Four of the seven sources arise after the data has been collected.

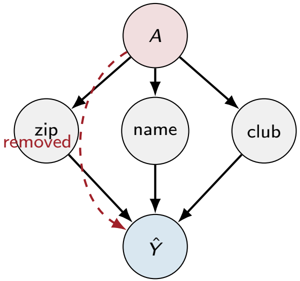
Feedback Loops: The Model Changes Its Own Training Data



- The loop is self-confirming: the prediction generates the evidence for itself.
- Same shape in recommenders (you only observe clicks on what you showed), in credit (you only observe repayment for approved applicants), and in hiring.
- **Detection:** hold out a slice from the policy, or log what the model *suppressed*, not just what it surfaced.

Fairness through unawareness: remove race, gender, age from the feature set and declare the model neutral.

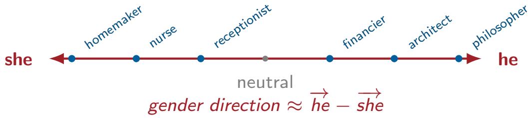
- Real features are **correlated** with the removed attribute: postcode, first name, school, device type, browsing history, purchase categories.
- A capable model *reconstructs* the attribute from the proxies. Removing the column removes your ability to **measure** the disparity, not the disparity itself.
- The Amazon reporting is exactly this: no gender field, gendered signal.
- Corollary: you often need the sensitive attribute — held securely and used only for auditing — in order to check for harm.



Cutting the direct edge leaves every indirect path intact.

Bias Lives in Learned Representations Too

Word embeddings trained on news text place words in a space where a **gender direction** is recoverable, and gender-neutral occupation words project onto it.



- Bolukbasi et al. showed the analogy solver completes *man : computer programmer :: woman : X* with "homemaker," and proposed a projection-based debiasing method.
- **Caveat worth teaching:** later work (Gonen & Goldberg, 2019) showed such projection mainly *hides* the bias — debiased vectors still cluster by stereotype, so the association is recoverable.
- Practical reading: a fix that makes a metric go down has not necessarily made the system fair.

Schematic drawn for this slide; the gender direction, the analogy result and the six occupation words (from the paper's extreme "she" / "he" lists) are from Bolukbasi, Chang, Zou, Saligrama & Kalai, "Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings," NeurIPS 2016, arXiv:1607.06520v1, Figure 1. Follow-up: Gonen & Goldberg, "Lipstick on a Pig," NAACL 2019, arXiv:1903.03862v2.

Turning "Fair" into Something You Can Measure

Notation for this section

- A — the group attribute (e.g. $A \in \{\mathrm{A}, \mathrm{B}\}$).
- $Y \in \{0, 1\}$ — the true outcome (repaid the loan, re-offended, was ill).
- $R \in [0, 1]$ — the score the model outputs.
- $\hat{Y} = \mathbf{1}[R > t]$ — the decision after thresholding.

Almost every group-fairness definition is one of three statements:

Family	**Statement**	**Plain reading**
Independence	$\hat{Y} \perp A$	the decision rate is the same for every group
Separation	$\hat{Y} \perp A \mid Y$	the error rates are the same for every group
Sufficiency	$Y \perp A \mid R$	a score means the same thing for every group

Framing after S. Barocas, M. Hardt and A. Narayanan, *Fairness and Machine Learning: Limitations and Opportunities*, MIT Press, 2023, fairmlbook.org, Chapter 3.

The Object Everything Is Computed From

One confusion matrix per group. Every definition below is a constraint tying two cells of these tables together across groups.

	$\hat{Y} = 1$	$\hat{Y} = 0$
$Y = 1$	**TP**	**FN**
$Y = 0$	**FP**	**TN**

Group A

- **Selection rate** $= \Pr\left(\hat{Y} = 1\right)$
- **TPR / recall** $= \Pr\left(\hat{Y} = 1 \mid Y = 1\right)$
- **FPR** $= \Pr\left(\hat{Y} = 1 \mid Y = 0\right)$
- **FNR** $= 1 - \mathrm{TPR}$
- **PPV / precision** $= \Pr\left(Y = 1 \mid \hat{Y} = 1\right)$
- **Prevalence** $p = \Pr(Y = 1)$

A fairness *audit* is: build this table per group, then compare a chosen cell across groups. The hard part is choosing which cell.

Definition 1: Demographic Parity (Independence)

Demographic parity / statistical parity

$$\Pr\left(\hat{Y} = 1 \mid A = \mathrm{A}\right) = \Pr\left(\hat{Y} = 1 \mid A = \mathrm{B}\right)$$

The model selects the same *proportion* of each group.

When it is the right question

- Allocating an opportunity where you do not trust the label (advertising, outreach, shortlisting).
- You believe the historical outcome Y is itself contaminated.
- Legal context: the US "four-fifths rule" in employment screening flags a selection-rate ratio below 0.8 as evidence worth investigating.

Where it goes wrong

- It ignores Y entirely. A model can satisfy it by selecting qualified people from one group and arbitrary people from the other.
- If the true base rates genuinely differ, enforcing it forces errors.
- Measured as a **ratio** (disparate impact) or a **difference** — report which.

Definition 2: Equalised Odds (Separation)

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Equalised odds — Hardt, Price and Srebro (2016)

$$\Pr\left(\hat{Y} = 1 \mid Y = y,\, A = \mathrm{A}\right) \;=\; \Pr\left(\hat{Y} = 1 \mid Y = y,\, A = \mathrm{B}\right) \qquad \text{for } y \in \{0, 1\}$$

Equal **TPR** and equal **FPR** across groups.

Equal opportunity — the one-sided version

Only $\Pr\left(\hat{Y} = 1 \mid Y = 1\right)$ must match: among people who *would* have succeeded, each group has the same chance of being selected.

- This is the ProPublica criterion: their complaint was unequal FPR and FNR.
- Use it when the burden of a **mistake** is what matters — a wrongly denied loan, a wrongly flagged defendant, a missed diagnosis.
- It conditions on Y, so it inherits whatever bias is in the label. If Y is "was arrested," equalised odds equalises errors against a biased ground truth.

Definition 2: Equalised Odds (Separation) (cont.)

M. Hardt, E. Price and N. Srebro, "Equality of Opportunity in Supervised Learning," NeurIPS 2016, arXiv:1610.02413v1.

Definition 3: Calibration and Predictive Parity (Sufficiency)

Calibration within groups

$$\Pr(Y = 1 \mid R = r,\, A = \mathrm{A}) \;=\; \Pr(Y = 1 \mid R = r,\, A = \mathrm{B}) \;=\; r$$

A score of 0.7 means a 70% chance for everyone, whatever their group. **Predictive parity** is the thresholded version: equal PPV across groups.

- This is the criterion the COMPAS vendor pointed to, and it is a genuinely important one: without it, a human reading the score is being misled about what it means.
- Well-trained probabilistic models are often *approximately* calibrated within groups by default — which is precisely why the conflict below is unavoidable rather than exotic.
- Check it with a **reliability diagram per group**, not with a single number.

You Cannot Have All Three

Chouldechova (2017), Equation (2.6)

$$\text{FPR} = \frac{p}{1-p} \cdot \frac{1-\text{PPV}}{\text{PPV}} \cdot (1-\text{FNR})$$

with p the prevalence $\Pr(Y=1)$ in that group.

Fix PPV to be equal across two groups. If $p_\text{A} \neq p_\text{B}$, the right-hand side differs, so FPR and FNR **cannot both** be equal. Predictive parity and error-rate balance are incompatible whenever base rates differ.

- Kleinberg, Mullainathan and Raghavan (2016) prove the same tension for calibrated *scores*: except in degenerate cases (equal base rates, or a perfect predictor), calibration and balance for both classes cannot hold together.
- Nothing about this depends on the model, the features, or the amount of data. It is arithmetic.

You Cannot Have All Three (cont.)

A. Chouldechova, "Fair prediction with disparate impact," *Big Data* 5(2), 2017, arXiv:1703.00056v1. J. Kleinberg, S. Mullainathan and M. Raghavan, "Inherent Trade-Offs in the Fair Determination of Risk Scores," ITCS 2017, arXiv:1609.05807v2.

Seeing the Conflict on a Toy Example

Two groups, 100 people each. Prevalence $\Pr(Y=1)$ in the population: $p_{\mathrm{A}} = 0.5$, $p_{\mathrm{B}} = 0.3$. A model that is **equally accurate within each group** (TPR $= 0.8$, TNR $= 0.8$ for both):

	$Y=1$	$Y=0$	TP	FP	**PPV**	**FPR**
Group A ($p = 0.5$)	50	50	40	10	$40/50 = \mathbf{0.80}$	$10/50 = \mathbf{0.20}$
Group B ($p = 0.3$)	30	70	24	14	$24/38 = \mathbf{0.63}$	$14/70 = \mathbf{0.20}$

- Error rates are balanced (equal TPR and FPR) — but a positive prediction means something different in each group (0.80 vs 0.63).
- Push PPV to match and the error rates separate. There is no threshold that fixes both.
- **Also note:** the selection rate is $50/100$ vs $38/100$, so demographic parity fails as well.

Group Fairness Is Not the Only Framing

Individual fairness

"Similar individuals should receive similar predictions" (Dwork et al., 2012).

- Attractive, and it catches cases group parity misses.
- Requires a task-specific **similarity metric** — which is the whole problem, moved somewhere else.

Causal / counterfactual fairness

"Would the decision change if this person had belonged to another group, holding everything else fixed?" (Kusner et al., 2017).

- Directly expresses what most people mean by discrimination.
- Needs a causal graph you are willing to defend, and the attribute is often not manipulable in the required sense.

For an introductory course: use group metrics because you can compute them, and remember that they are a screening tool, not a certificate.

So Which Definition Do You Use?

This is a decision about **harms**, not a hyperparameter search. Four questions, in order:

1. **Who is affected, and what is the harm?** Being wrongly *denied* something (allocative) or being wrongly *flagged* (punitive) point to different error rates.

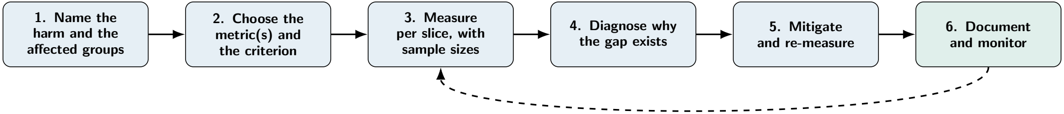
2. **Do you trust the label Y?** If the label encodes past discrimination, definitions that condition on Y inherit it — lean towards independence.

3. **Is the score read by a human?** If yes, calibration within groups is close to mandatory, or you are misinforming the decision-maker.

4. **What does the law in your jurisdiction require?** Sometimes the answer is fixed for you — and per-group thresholds may be illegal (see the next section).

Write the chosen criterion, and the criteria you consciously gave up, into the model card. An unstated choice is still a choice.

The Audit Loop



- Step 4 is the one people skip. A gap caused by **too few samples** needs data; a gap caused by a **proxy label** needs a new label; a gap caused by a **threshold** needs a threshold. The fixes are not interchangeable.
- Step 6 is not optional: distributions move, and a model that was audited in March is not audited in December.

Disaggregated Evaluation: The Core Technique

One number for the test set, then the same number for every slice you can define.

Slice	n	**Selection rate**	**TPR**	**FPR**	**PPV**
Overall	10,000	0.24	0.78	0.11	0.71
Group A	8,200	0.26	0.81	0.10	0.73
Group B	1,800	0.14	0.62	0.13	0.58
Group B, age < 30	420	0.11	0.49	0.14	0.51
Group B, age ≥ 60	180	0.09	0.44	0.12	0.47

Illustrative numbers, constructed for this slide.

- The overall row is fine. Group B is not, and the intersectional rows are worse — the pattern from Gender Shades, reproduced in miniature.
- Always print n beside the metric. Always print the **worst slice** in the summary, not just the mean.
- In `scikit-learn`, this is a `groupby` plus `classification_report`; in `fairlearn` it is `MetricFrame`.

Four Things That Go Wrong in Practice

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- **Small slices are noisy.** A TPR of 0.44 estimated from 180 positive cases carries a 95% interval of about ± 0.07; from 40 cases, about ± 0.15. Report intervals, or you will chase noise and "fix" gaps that were never there. A bootstrap is fine here.
- **Multiple comparisons.** Slice by 8 attributes and some slice will look bad by chance. Pre-register the slices you care about; treat the rest as exploratory.
- **You may not have the group labels.** They are often unavailable by law or by policy. Options: a consented subset for audit only, an independently collected audit set, or proxy-based estimates — each with its own error, which must be stated.
- **Intersections multiply.** Four binary attributes give 16 cells and most will be tiny. Choose the intersections that correspond to a plausible harm, rather than enumerating everything.

Where Mitigation Can Be Applied

Pre-processing reweigh, resample, transform features	**In-processing** fairness constraint or penalty in the objective	**Post-processing** adjust thresholds or scores per group

- **Access decides the option.** Third-party model behind an API: post-processing only. Your own training loop: all three.
- **None of them repair a bad label or a missing population.** Those are data problems, and mitigation applied on top will only redistribute the error.

Pre-processing and In-processing

Pre-processing — change the data so the disparity is not there to be learned.

- **Reweighing** (Kamiran & Calders, 2012): weight each (group, label) cell so group and label become independent in the weighted sample. Works with any estimator taking `sample_weight`.
- **Resampling**: same effect, but duplicates or discards rows.
- **Targeted data collection**: usually the highest-value intervention, and the one most often skipped because it costs money rather than code.
- **Representation transforms**: learn an encoding from which A cannot be predicted.

Model-agnostic and done once — but you optimise a proxy, and the downstream model can partly undo it.

In-processing — put the requirement into the training objective.

- **Constrained optimisation**: minimise loss subject to $|\mathrm{TPR_A} - \mathrm{TPR_B}| \leq \varepsilon$. Agarwal et al. (2018) reduce this to a sequence of ordinary cost-sensitive fits, so any learner works — this is `fairlearn`'s `ExponentiatedGradient`.
- **Regularisation**: add $\lambda \cdot$ (disparity) to the loss; λ is an explicit dial on the trade-off.
- **Adversarial debiasing**: train the predictor against an adversary trying to recover A from the prediction or representation.

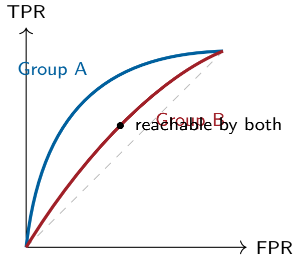
Enforced where the model is fitted — but needs training access, adds tuning, and holds on the *training* distribution.

F. Kamiran and T. Calders, *Knowledge and Information Systems* 33(1), 2012. A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford and H. Wallach, "A Reductions Approach to Fair Classification," ICML 2018, arXiv:1803.02453v3.

Post-processing: Group-Specific Thresholds

Hardt et al. (2016) show that if you keep the score R and choose a **separate threshold per group**, you can achieve equalised odds — and they give the optimal choice.

- Each group's ROC curve gives the achievable (FPR, TPR) pairs.
- Equalised odds requires landing on the **same point** for both groups, so the feasible set is the intersection.
- The best you can do is the intersection point that maximises utility — which is generally *worse* than either group's own optimum.



Schematic drawn for this slide, after the construction in Hardt et al. (2016).

Legal caveat

Applying a different threshold by protected group can itself be unlawful in some jurisdictions and domains — for example, adjusting employment-test scores by race is prohibited in the United States under the Civil Rights Act of 1991. Check before you ship; the statistically optimal fix is not always the permitted one.

Expect to give something up

- Constraining a metric usually costs some accuracy. Report the curve (disparity vs accuracy), not one point — it turns an argument into a decision.
- Sometimes the cost is near zero, which is worth discovering.
- “Levelling down” — degrading the strong group to close the gap — satisfies the metric and helps nobody. Say explicitly whether it is acceptable.

After deployment

- Re-run the disaggregated report on production traffic on a schedule.
- Alert on disparity, not only on accuracy drift.
- Keep a channel for people to contest a decision, and log contested cases — they are your best source of failure modes.

Libraries

- `fairlearn` (metrics, reductions, threshold optimiser)
- `AIF360` (a wide catalogue of algorithms)
- `aequitas` (audit reports)

Lab: `Fairness_Audit.ipynb` — disaggregate on UCI Adult, show the parity/odds tension, apply threshold post-processing, and measure what it cost.

Why Explain a Model At All?

- **Debugging.** By far the most common real use. Explanations expose leakage, shortcut features, and broken preprocessing that accuracy alone hides.
- **Trust and adoption.** A clinician or underwriter who cannot see the reasoning will not use the tool — or will over-trust it.
- **Recourse.** Telling a rejected applicant *what to change* is a different requirement from telling an engineer what the model used.
- **Regulation.** The EU AI Act gives affected persons a right to an explanation of individual decisions from certain high-risk systems (Article 86); GDPR Article 22 constrains solely automated decisions.
- **Science.** In a scientific application the model *is* the hypothesis, and what it uses is the result you wanted.

Each purpose demands a different artefact. Decide the purpose **before** choosing the method.

Three Axes for Sorting Every Method

Axis		
Intrinsic vs **post-hoc**	The model is readable as it stands: linear/logistic regression, a shallow tree, a rule list, a GAM.	An external procedure explains a fitted black box: permutation importance, SHAP, LIME, Grad-CAM.
Global vs **local**	What does the model do *in general*? Feature importances, partial dependence.	Why *this* prediction for *this* row? SHAP values, LIME, counterfactuals.
Model-specific vs **model-agnostic**	Uses the internals: tree paths, gradients, attention, TreeSHAP.	Only needs `predict()`: permutation importance, KernelSHAP, LIME.

Model-agnostic buys generality at the price of speed and, often, fidelity. Model-specific is faster and usually more faithful, but locks you to a model family.

Try an Interpretable Model First

Rudin (2019)

For high-stakes decisions, prefer a model that is interpretable by construction over a black box plus a post-hoc explanation — because the explanation is an **approximation** of the model, and an approximation can be wrong in exactly the cases you care about.

- On many tabular problems, a well-regularised logistic regression, a depth-4 tree, or a generalised additive model is within a point or two of a gradient-boosted ensemble. **Measure the gap before assuming you need the black box.**
- If the gap is real and worth it, take the black box — but then the explanation needs its own validation, exactly as the model does.
- Interpretability is not a property of the model alone. A linear model with 4,000 engineered features is not interpretable; a depth-3 tree over four meaningful features is.

C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence* 1:206–215, 2019, arXiv:1811.10154v3.

Recipe

1. Measure the model's score on the validation set.
2. Shuffle one feature column, breaking its relationship with the target.
3. Re-score. The **drop** is that feature's importance.
4. Repeat several times and average; then restore the column and move on.

Why start here

- Model-agnostic, ten lines of code, needs only `predict()`.
- Measures importance **for the metric you care about**.
- Computed on *held-out* data, so it reflects generalisation, not fit.

Where it misleads

- **Correlated features split the credit**, and each looks unimportant because the other covers for it.
- Shuffling creates impossible rows (age 8, income $200k), so you evaluate the model off its data manifold.
- It is global only — it says nothing about a single prediction.

A Trap: Built-in Tree "Feature Importance"

`RandomForest.feature_importances_` is **mean decrease in impurity** (MDI), accumulated on the *training* data.

- It is biased towards **high-cardinality** features: a continuous feature or a random ID column offers many split points, so it looks informative even when it is pure noise.
- It is computed on training data, so it rewards features the model used to overfit.
- Classic demonstration: add a column of random integers to your dataset and it can outrank real predictors in MDI — while permutation importance on held-out data correctly gives it roughly zero.

Rule: never report `feature_importances_` as "the model's feature importance" without saying which definition you used. Prefer permutation importance or SHAP on held-out data.

SHAP: Attribution as a Fair Payout

The analogy. A team of players cooperates and earns a payout. How much of it does each player deserve? Here the "players" are the features of one row, and the "payout" is $f(x) - \mathbb{E}[f(X)]$: how far this prediction sits from the average prediction.

Shapley value (Shapley, 1953)

Consider adding feature i to a coalition S of features already present, and record the change in the prediction. The Shapley value ϕ_i is the **average of that change over all possible orderings** in which features could be added:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \Big[v(S \cup \{i\}) - v(S) \Big]$$

- The averaging over orderings is what makes it symmetric: no feature is privileged by being "first".
- There are $2^{|N|}$ subsets, so the exact computation is exponential. Everything practical in SHAP is an efficient approximation of this quantity.

Why This Particular Formula: The Axioms

Lundberg and Lee (2017) define a class of *additive feature attribution* methods and show that within it exactly **one** solution satisfies three properties — the Shapley values. They also show that several earlier methods (LIME with a particular kernel, DeepLIFT, layer-wise relevance propagation) are members of that same class.

- **Local accuracy (efficiency).** The parts add up to the whole:

$$f(x) = \underbrace{\mathbb{E}[f(X)]}_{\text{base value}} + \sum_i \phi_i$$

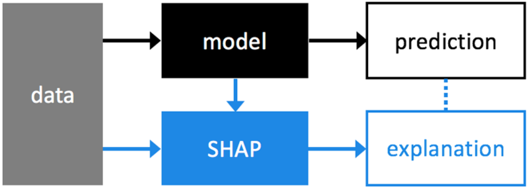
  This is what makes a SHAP plot readable: the bars literally sum to the prediction.

- **Missingness.** A feature that is absent from the model gets $\phi_i = 0$.
- **Consistency.** If you change the model so that a feature's marginal contribution never decreases, its attribution cannot decrease. Common tree importance measures (gain, split count) violate this — the formal reason to prefer SHAP for tree models.

Why This Particular Formula: The Axioms (cont.)

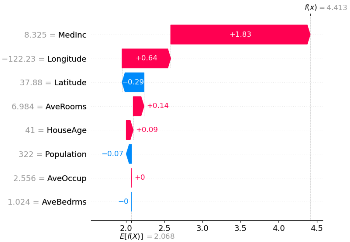
S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” NeurIPS 2017, arXiv:1705.07874v2.

What SHAP Consumes and Produces



Source: `shap` library documentation artwork, github.com/shap/shap (`docs/artwork/shap_diagram.png`).

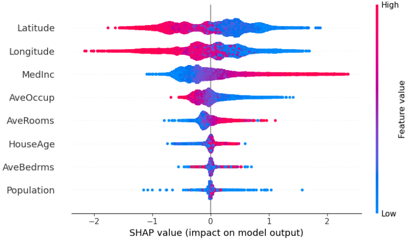
- SHAP does not change the model. It takes the **fitted model** plus a **background dataset** and produces one attribution per feature per row.
- The background dataset defines $\mathbb{E}[f(X)]$ — the reference the explanation is measured against. **Change the background and every number changes**, so record which rows you used.
- A useful phrasing for stakeholders: *"compared with a typical applicant, this applicant's income pushed the score up by 0.31."*



Source: shap library documentation artwork, github.com/shap/shap (docs/artwork/california_waterfall.png); model trained on the California Housing dataset, target in units of $100k.

- Read bottom to top: start at the base value $\mathbb{E}[f(X)] = 2.068$, add each feature's contribution, arrive at this row's prediction $f(x) = 4.413$.
- The grey number on the left is the feature's actual value for this row.
- This is the artefact to show a domain expert. One row, one story, and it adds up.

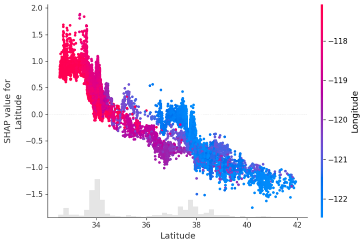
Global View: The Beeswarm Summary Plot



Source: `shap` library documentation artwork, github.com/shap/shap (`docs/artwork/california_beeswarm.png`).

- One dot per row per feature; horizontal position is the SHAP value, colour is the **feature value** (red high, blue low). Features are ordered by mean $|\phi|$.
- Read the **direction**: for `MedInc`, red dots sit on the right, so higher median income pushes the predicted price up — monotone and sensible.
- Read the **spread**: a feature with a wide swarm matters a lot for some rows and not at all for others. A bar chart of mean $|\phi|$ would have hidden that.

Interaction View: The Dependence / Scatter Plot



Source: shap library documentation artwork, github.com/shap/shap (docs/artwork/california_scatter.png).

- *x*-axis: the feature's value. *y*-axis: its SHAP value. This is the model's learned response to that feature, without assuming it is linear or monotone.
- Colouring by a second feature exposes **interactions**: here the vertical spread at a given latitude is explained by longitude — the model has learned geography, not a one-dimensional rule.
- Sharp steps are informative: they are the split points of the underlying trees.

Which SHAP Estimator?

Estimator	Applies to	Notes
TreeSHAP	Trees, random forests, gradient boosting (XGBoost, LightGBM, CatBoost)	Exact, in polynomial time, by exploiting the tree structure. Fast enough for a whole dataset. **Use this whenever you can.**
KernelSHAP	Anything with `predict()`	Weighted local linear regression over sampled coalitions. Model-agnostic but **slow**; sample a few hundred background rows and a subset of instances.
DeepSHAP / Gradient	Neural networks	Uses backpropagation and a DeepLIFT-style approximation; much cheaper than KernelSHAP.
Linear	Linear models	Closed form: $\phi_i = w_i(x_i - \mathbb{E}[X_i])$ under feature independence.

Polynomial-time exact tree attribution: S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence* 2:56–67, 2020.

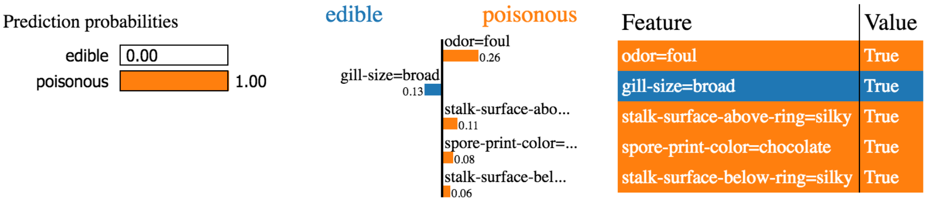
LIME: Fit a Simple Model in a Small Neighbourhood

Pick the row x to explain → Perturb it: sample many nearby variants → Ask the black box to label them → Weight samples by closeness to x → Fit a sparse linear model on the weights

- The coefficients of that little linear model *are* the explanation.
- "Perturb" is domain-specific: for text, delete words; for images, switch superpixels on and off; for tabular data, sample around the row.
- The surrogate is deliberately **sparse** (a handful of features) so a human can read it. Complexity is traded for legibility, on purpose.

M. T. Ribeiro, S. Singh and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," KDD 2016, arXiv:1602.04938v3.

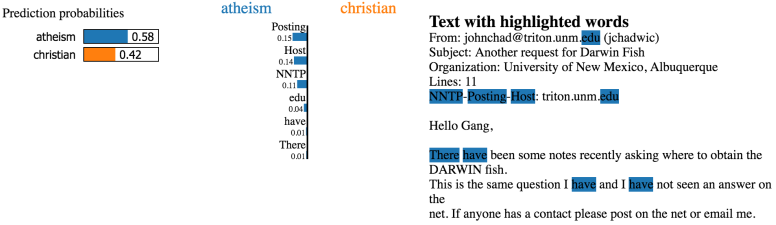
LIME on Tabular Data



Source: `lime` library documentation, github.com/marcotcr/lime (`doc/images/tabular.png`); mushroom edibility classifier.

- Left: the predicted class probabilities. Middle: signed contributions of the top features. Right: the actual feature values for this row.
- The explanation is in terms of **discretised bins** (`odor=foul`), which is why it reads well to a non-specialist — and why the numbers depend on how you binned.
- Note the contrast with SHAP: LIME's weights are coefficients of a local surrogate. They do **not** sum to the prediction.

LIME on Text: Finding a Shortcut



Source: `lime` library documentation, github.com/marcotcr/lime (`doc/images/twoclass.png`); 20 Newsgroups atheism-vs-christianity classifier.

- The classifier reports 0.58 for "atheism." The explanation shows the evidence: `Posting`, `Host`, `NNTP`, `edu` — **email header artefacts**, not content.
- This is the single most valuable use of local explanations: a high-accuracy model that is right for a reason that will not survive contact with new data.
- You would never have found it from the accuracy score.

(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception network, highlighting positive pixels. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

Source: Figure 4 of M. T. Ribeiro, S. Singh and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," KDD 2016 (arXiv:1602.04938v3), as reproduced in the `lime` repository.

- The image is segmented into superpixels; perturbation switches groups of them off.
- Each panel shows the superpixels supporting a *different* predicted class — the explanation is always **relative to a target class**, never absolute.
- Useful check: if the supporting region is the background rather than the object, you have found a dataset artefact.

Pitfalls You Must Know Before Using Either

أكاديمية كاوست
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- **Correlated features.** Both methods perturb features independently, producing rows that could not exist. Attribution then gets spread arbitrarily among correlated features, and can be assigned to a feature the model barely uses. Group correlated features and attribute to the group.

- **The background / neighbourhood is a choice.** SHAP values are relative to the background dataset; LIME weights depend on the kernel width and the perturbation scheme. Report what you used — otherwise the numbers are not reproducible.

- **Instability.** Re-running LIME with a different seed can reorder the top features. Always run it several times and check the explanation is stable before you show it to anyone.

- **Explanations can be attacked.** Slack et al. (2020) built classifiers that behave in a biased way on real data but detect the synthetic perturbation distributions used by LIME and SHAP, and answer innocuously there — so the explanation hides the bias.

Pitfalls You Must Know Before Using Either (cont.)

Lab: `Model_Interpretability_SHAP_LIME.ipynb` — SHAP (Tree and Kernel) and LIME on the same tabular model, with an explicit demonstration of instability under correlated features.

D. Slack, S. Hilgard, E. Jia, S. Singh and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods," AIES 2020, arXiv:1911.02508v2.

Explaining the Model vs Explaining the World

What SHAP answers

"How does *this fitted function* distribute its output across the inputs I gave it?"
A statement about the model. Correct as far as it goes.

What people hear

"Which factors *cause* this outcome, and what should this person change?"
A causal statement. Not supported by the attribution.

- Kumar et al. (2020) argue that the mathematical appeal of Shapley values does not transfer to the human goals people bring to explanations, and that closing the gap requires causal assumptions the method does not make.
- If the user needs **recourse** ("what do I change?"), the right tool is a **counterfactual explanation** — the nearest feasible input that flips the decision — not a feature attribution.
- **Not unique, and not a defence.** Different methods, backgrounds and seeds give different attributions for the same prediction; a reviewer finding an explanation satisfying is not evidence the model is correct.
- **Rule of thumb:** use explanations to **generate hypotheses**, then test each one by intervention — an ablation, a re-fit without the feature, a fresh slice.

I. E. Kumar, S. Venkatasubramanian, C. Scheidegger and S. Friedler, "Problems with Shapley-value-based explanations as feature importance measures," ICML 2020, arXiv:2002.11097v2.

Why Pixels and Tokens Are Harder

- There is no meaningful "feature" called pixel $(117, 204)$. Attribution must be summarised over **regions**, or it is unreadable.
- There are hundreds of thousands of inputs, so coalition-based methods are far too expensive without a gradient shortcut.
- The inputs are strongly correlated by construction — neighbouring pixels, adjacent tokens — which is exactly the regime where independent perturbation misleads.

Vanilla gradient saliency (Simonyan, Vedaldi & Zisserman, 2014)

Take the score $s_c(x)$ for class c, backpropagate to the input, and display $\left|\partial s_c / \partial x\right|$ as an image.

- Interpretation: the pixels whose small changes would move the class score most. In practice raw gradients are **visually noisy** — speckle rather than an object.
- Common repairs: **SmoothGrad** (average the gradient over noisy copies of the input) and **Integrated Gradients** (accumulate gradients along a path from a baseline to the input, restoring an additivity property analogous to SHAP's).

Why Pixels and Tokens Are Harder (cont.)

- Each needs a **baseline** or a noise level. As with SHAP's background, the choice changes the picture.

K. Simonyan, A. Vedaldi and A. Zisserman, "Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps," ICLR 2014 Workshop, arXiv:1312.6034v2. M. Sundararajan, A. Taly and Q. Yan, "Axiomatic Attribution for Deep Networks," ICML 2017, arXiv:1703.01365v2.

The idea, in three steps

1. Take the feature maps of the **last convolutional layer** — they still carry spatial position, but encode semantics.
2. Weight each map by the average gradient of the target class score with respect to that map ("how much does this channel matter for this class").
3. Sum the weighted maps, apply ReLU to keep positive evidence, and upsample to the image size.

- Coarse but **class-discriminative**: change the target class and the heatmap moves.
- Works on any CNN without retraining; ViT variants exist.
- **Practitioner's use:** check that the heatmap sits on the object, and not on a watermark, a ruler, a hospital tag, or the background.

Same image, same network, two target classes: *dog* (left), *cat* (right).

Source: example outputs from the `pytorch-grad-cam` library (ResNet-50). Method: R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh and D. Batra, "Grad-CAM," ICCV 2017, arXiv:1610.02391v3.

Sanity Checks: A Pretty Heatmap Is Not Evidence

Adebayo et al. (2018)

Two tests any saliency method should pass:

- **Model randomisation:** progressively randomise the trained weights. The explanation should change.
- **Data randomisation:** retrain on randomly permuted labels. The explanation should change.

They find that **some existing saliency methods are independent both of the model and of the data generating process** — producing similar-looking maps regardless.

- Such a method is closer to an **edge detector** than to an explanation. It looks convincing because the object outline is where the edges are.
- Practical consequence: run the randomisation test yourself on the method you plan to ship. It costs one retraining run.

J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt and B. Kim, "Sanity Checks for Saliency Maps," NeurIPS 2018, arXiv:1810.03292v3.

“Attention Is Not Explanation” — and the Reply

أكاديمية كاوست
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An **attention** layer assigns a weight to every input token, and those weights are routinely displayed as “what the model looked at.”

Jain & Wallace (NAACL 2019)

Across several NLP tasks:

- Attention weights are frequently **uncorrelated** with gradient-based importance.
- One can construct **very different** attention distributions that give **the same** prediction.

Conclusion: standard attention modules do not provide meaningful explanations.

Wiegreffe & Pinter (EMNLP 2019)

The conclusion depends on what you mean by “explanation.” They propose four tests — including a uniform-weight baseline and a diagnostic on frozen weights — and find that adversarial attention distributions, where they exist, do poorly on the diagnostic.

Attention is not *no* evidence; it is not *sufficient* evidence.

What to do in practice

Use attention maps as a **debugging view**, never as a justification for a user or a regulator. For a transformer attribution, use a method built for it (Integrated Gradients, attention rollout, gradient $\times$ input) and validate it.

S. Jain and B. C. Wallace, “Attention is not Explanation,” NAACL 2019, arXiv:1902.10186v3. S. Wiegreffe and Y. Pinter, “Attention is not not Explanation,” EMNLP 2019, arXiv:1908.04626v2.

Documentation Is Part of the Model

A model that ships without documentation forces every downstream user to rediscover its limits — usually in production.

Model card

Travels **with the trained model**. Answers: what is it for, who is it for, where does it fail, and how well does it work *per group*.

Mitchell et al., FAT* 2019.

Datasheet

Travels **with the dataset**. Answers: why was it collected, from whom, with what consent, how was it cleaned, and what should it not be used for.

Gebru et al., CACM 2021.

Both are borrowed ideas

A model card is a **datasheet for a component**, as in electronics; a dataset datasheet is the same idea one step upstream. Neither is a research artefact — they are engineering hygiene, and the EU AI Act turns much of their content into a legal obligation for high-risk systems.

What Goes in a Model Card

Section	Content
Model details	Owner, date, version, model type, training algorithm, licence, citation, contact.
Intended use	Primary intended uses and users; **out-of-scope** uses stated explicitly.
Factors	Groups, instrumentation, and environments the performance is expected to vary over.
Metrics	Which measures, which decision threshold, and how uncertainty was estimated.
Evaluation data	Which datasets, why they were chosen, and how they were preprocessed.
Training data	Distribution over the same factors, as far as it can be disclosed.
Quantitative analyses	Results **unitary** (per factor) and **intersectional** (per combination).
Ethical considerations	Sensitive data, risks to people, mitigations applied, known problematic uses.
Caveats & recommendations	Untested groups, open concerns, what testing should happen next.

M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji and T. Gebru, "Model Cards for Model Reporting," FAT* 2019, Section 4, arXiv:1810.03993v2.

A Filled Example (1/2) — Invented for Teaching

Model card — `loan-default-v3`

Model details. Gradient-boosted trees (LightGBM, 400 trees, depth 6). Owner: Credit Risk team. Version 3.1, trained 2026-05-14 on applications from 2021–2025. Contact: `risk-ml@example.org`.

Intended use. Produces a default-probability score used to *prioritise manual review* of personal-loan applications between \$1k and \$25k in Market X. Primary users: underwriting analysts.

Out of scope. Not for automatic decline without human review. Not for business loans, mortgages, or other markets. Not for pricing. Not valid for applicants under 21 (fewer than 300 in training).

Factors. Reported over: age band, gender, region, income decile, thin-file vs thick-file credit history, and the region × thin-file intersection.

Metrics. AUC and recall at the operating threshold 0.31, chosen to keep the manual-review queue at 20% of volume. 95% intervals from 1,000 bootstrap resamples.

Fairness criterion adopted. Equal opportunity on the review decision (equal TPR across region and thin-file status). Demographic parity was **not** targeted, because base default rates differ materially by region; this trade-off was signed off by the Model Risk Committee on 2026-05-02.

Model card — `loan-default-v3` (continued)

Quantitative analyses.

Slice	n	AUC	TPR @ 0.31	FPR @ 0.31
Overall	42,300	0.81	0.74	0.18
Region North	31,500	0.82	0.76	0.17
Region South	10,800	0.77	0.66	0.21
Thin file	6,100	0.71	0.58	0.24
South $\times$ thin file	1,450	**0.68**	**0.52**	**0.27**

Ethical considerations. Region correlates with an attribute protected in Market X. Region is not an input, but postcode-derived features are; a proxy audit is run quarterly. A wrongly high score delays access to credit, so recall on the worst slice is the primary fairness metric.

Caveats and recommendations. The worst slice is 3.4% of volume and is **materially worse** than overall — route it to a specialist queue rather than the standard one. Not validated after a policy change to income verification (2026-03). Re-audit on 2026-11-14 or after any change to the acquisition channel.

Datasheets for Datasets

Seven groups of questions, answered by whoever created the dataset:

1. **Motivation.** Why was it created, by whom, funded by whom?
2. **Composition.** What is an instance? How many? What is missing? Does it contain data about people, or confidential/sensitive content?
3. **Collection process.** How was it acquired, by whom, over what period? Were the people informed, and did they consent?
4. **Preprocessing / cleaning / labelling.** What was done, and is the raw version still available?
5. **Uses.** What has it been used for; what should it **not** be used for?
6. **Distribution.** Who gets it, under what licence, with what restrictions?
7. **Maintenance.** Who maintains it, how are errors fixed, is there versioning, is there a retention limit?

The questions are deliberately uncomfortable. "We do not know" is a valid answer — and is itself the finding.

Where these live. Hugging Face model and dataset cards are exactly this format. Keep the card in the *same repository as the training code*, generate its quantitative table *from the evaluation script*, and make it a merge requirement — a hand-copied table is stale within a week.

Datasheets for Datasets (cont.)

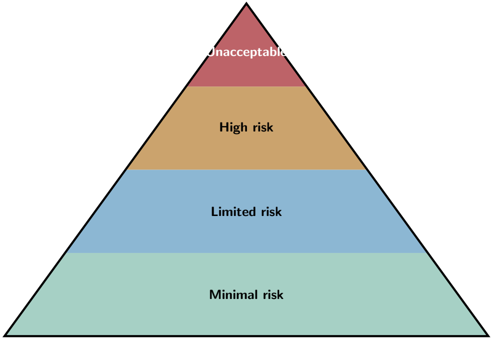
Exercise (15 minutes)

Take the last model you trained. Write its **intended use**, its **out-of-scope uses**, and one **quantitative table with slices**. Most people discover they cannot fill in the second one.

T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III and K. Crawford, "Datasheets for Datasets," *Communications of the ACM* 64(12):86–92, 2021, arXiv:1803.09010v8.

The EU AI Act: Risk Tiers

Good practice becomes a **requirement** once a system falls into a regulated tier. Orientation, not legal advice.



Prohibited (Art. 5), in part. Social scoring; untargeted scraping of facial images to build recognition databases; emotion inference at work or school, except for medical or safety reasons; manipulative or exploitative systems.

Permitted with obligations. Employment, credit, education, essential services, law enforcement, biometrics, and safety components of regulated products (Annexes I and III).

Transparency duties (Art. 50). Tell people they are interacting with an AI system; label synthetic media.

No specific obligations. Spam filters, low-stakes recommendation, most internal tooling — where most course projects sit.

Source: Regulation (EU) 2024/1689 (Artificial Intelligence Act), artificialintelligenceact.eu.

The EU AI Act: What a High-Risk Provider Must Produce

- A **risk management system** maintained across the lifecycle.
- **Data governance**: relevant, representative, and examined for bias.
- **Technical documentation** and automatic **logging**.
- **Instructions for use** that state accuracy, limitations, and intended conditions.
- Effective **human oversight** by design.
- Appropriate **accuracy, robustness and cybersecurity**.
- **Post-market monitoring** and **serious-incident reporting**.

Recognise anything?

Disaggregated bias evaluation, documented intended use, stated limitations, logged decisions, monitoring.
That is the model card and the audit from the last two sections, made mandatory. The work does not change; the audience does.

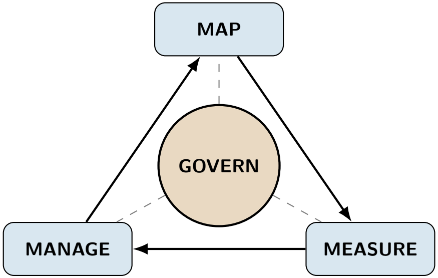
Penalties (Article 99)

Up to **EUR 35M or 7%** of worldwide annual turnover for prohibited practices; **EUR 15M or 3%** for most other operator obligations; **EUR 7.5M or 1%** for supplying incorrect information, in each case whichever is *higher*. For SMEs and start-ups the same figures apply but the rule flips to whichever is *lower*.

The EU AI Act: What a High-Risk Provider Must Produce (cont.)

Regulation (EU) 2024/1689, Chapter III and Article 99. Entered into force 1 August 2024; prohibitions and AI-literacy duties applicable from 2 February 2025; general-purpose AI model obligations from 2 August 2025. Regulation (EU) 2026/1744 (the "AI Omnibus," in force 27 July 2026) deferred the high-risk obligations: Annex III to 2 December 2027, Annex I to 2 August 2028, and softened Article 4 to a duty to *support the development of* AI literacy. These dates have already moved once — check the consolidated text.

NIST AI Risk Management Framework



- **Govern** — policies, roles, accountability. Cuts across the other three rather than sitting in sequence with them.
- **Map** — establish the context: who is affected, what could go wrong, what is the system actually for.
- **Measure** — test it: metrics, disaggregated evaluation, red-teaming, documented uncertainty.
- **Manage** — prioritise, mitigate, monitor, and decide when to stop.

Voluntary, sector-neutral, and widely used as the internal scaffolding for an AI governance programme — it tells you *how* to organise risk management, where the AI Act tells you *what* you must do. ISO/IEC 42001:2023, *Information technology — Artificial intelligence — Management system*, plays a similar role as a certifiable management-system standard. A Generative AI Profile (NIST AI 600-1, 2024) adapts the framework to generative systems.

Source: NIST, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, January 2023, nist.gov.

Organisational Practice

Review before release

- A named reviewer who is **not** the model's author signs off the card and the disaggregated results.
- Scale the review to the stakes: a recommender for internal search does not need a committee; a credit model does.
- Record the decision *and the trade-off accepted*, so it can be revisited.

Red-teaming

- Deliberately try to make the system fail for a specific group or input type, before users do.
- Give the red team the failure taxonomy from this lecture as a starting checklist.

After release

- **Incident reporting.** A defined path for "the model did something harmful," with an owner and a deadline — not a support ticket queue.
- **Contestability.** A person affected by a decision can ask for it to be reviewed by a human, and the request is logged.
- **Scheduled re-audit.** Put the date in the model card and treat it like a certificate expiry.
- **A kill switch.** Know in advance how you turn the model off and what the fallback is.

The common failure is not the absence of a policy. It is a policy with no owner and no date.

Agents and LLMs: Covered in the NLP Course

Everything in this lecture applies to a model that maps an input to a prediction. Systems that **take actions** — call tools, browse, write files, spend money — add a distinct threat model that this deck deliberately does not duplicate.

NLP course: AI Safety for Agents

- Prompt injection, direct and indirect.
- Excessive agency and tool misuse.
- Sandboxing, least privilege, and human-in-the-loop gates.
- Agent-specific benchmarks and red-teaming.
- OWASP Top 10 for LLM Applications.

What carries over

- Disaggregated evaluation — an agent can fail more often for some users or some languages.
- Documentation of intended and out-of-scope use.
- The same governance frameworks: the EU AI Act and NIST AI RMF cover both, and the AI Act adds obligations for general-purpose AI models.

If you are shipping an agent, take both lectures. Neither is a substitute for the other.

What to Take Away

- **Aggregate metrics hide subgroup failure.** The single most valuable habit from this lecture is the disaggregated evaluation table, with sample sizes, including intersections.
- **Bias is mostly a specification problem.** It enters through the target definition, the label, the sampling frame, and the deployment context — not only through "the data." Dropping the sensitive attribute does not remove it.
- **The fairness definitions genuinely conflict.** When base rates differ, calibration and error-rate balance cannot both hold. Choose the criterion that matches the harm, and write down what you gave up.
- **Explanations describe the model, not the world.** SHAP and LIME are excellent for debugging, sensitive to correlated features and to the background distribution, and not causal. Validate a hypothesis before acting on it.
- **Documentation is the deliverable.** The model card and the audit are what a reviewer, a customer, or a regulator will actually read — and under the EU AI Act, much of their content is now required for high-risk systems.

A Checklist You Can Use on Monday

Before training

1. Write one sentence: who is affected, and how could they be harmed?
2. Ask what your label *actually* measures, and whether that differs by group.
3. Check the composition of your data against the intended use population.
4. Fit an interpretable baseline and record its score.

Before shipping

5. Produce the disaggregated table with n and intervals.
6. Name the fairness criterion and what you traded away.
7. Run SHAP or permutation importance and look for a shortcut feature.

At release

8. Write the model card, including out-of-scope uses and the worst slice.
9. Get a review from someone who did not build it.
10. Set the re-audit date and the monitoring alert on disparity.

Items 1, 2 and 5 catch most of what goes wrong, and none of them require a new library.

Labs: `Model_Interpretability_SHAP_LIME.ipynb` and `Fairness_Audit.ipynb`.

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